

ASILIX

Software Requirements Specification - SRS

Safety Services & Licensing Management System



1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the functional and non-functional requirements for the Safety Services & Licensing Management System — a centralized cloud (SaaS) platform that manages the operations of technical safety and licensing companies. It governs licensing-request processing, automated activity/area-based routing, centralized field execution and procurement, document and signature authentication, and per-tenant isolation.

1.2 Product Scope

The platform provides an isolated workspace per subscribed safety company (tenant), an automated rule engine that classifies licensing requests by activity type and facility area, a centralized operations and procurement portal where the operations officer records all field activity (material invoices, field labor governance, technical inspection reports), and an interactive client portal for service selection, facility data entry, file uploads, and live tracking by Job Number — protecting the company's financial and legal records. In the current MVP, all field operations are performed through the responsive web browser portal rather than a native mobile application.

2. Overall Description

2.1 Product Perspective

The system is an integrated operational and administrative environment connecting client-facing portals with internal control panels for sales agents, consulting engineers, and the central operations & projects officer. All interfaces synchronize in real time to exchange data and update request, procurement, and obstacle-log states simultaneously.

2.2 Product Functions

At a high level, the system will:

- Provision and isolate a dedicated tenant environment, custom subdomain, and branding per subscribed company.
- Authenticate users with TOTP multi-factor authentication and role-based access control.
- Capture client intake (CR/700 number, built-up area, GPS, documents, blueprints).
- Automatically route requests by area thresholds and ISIC activity hazard bands.
- Manage consulting workspace, blueprint review, and three specialized technical reports.

- Run centralized web-based field execution: project silos, procurement/VAT, labor governance, obstacle logs, signature capture, recurring maintenance scheduling.
- Provide tracking, internal chat, profitability analytics, BI dashboards, and SLA scorecards.
- Enforce non-functional security, performance, and integration requirements.

2.3 User Classes and Characteristics

- **Super Admin:** Holds top-level authority to manage subscribed safety companies, review packages, and monitor overall platform performance.
- **Company Admin:** Controls their own safety company's options, manages staff accounts, and views profitability analytics and operational/financial reports.
- **Consulting Engineer:** Receives new licensing/contract requests, reviews uploaded data, studies blueprints, manages and generates quotes, assigns inspection agents, and prepares/approves the three specialized technical reports.
- **Operations & Projects Officer:** The pivotal field party; receives work orders on approval, manages item procurement, records tax invoices, schedules and enters field labor data and wages, documents

inspection/obstacle reports, and closes tickets with client signature/stamp photos.

- **Sales Agent:** Adds target clients in the field and tracks the sales pipeline from request to approval, with commission isolation.
- **Client:** Uses services by filling forms, reviewing quotes, and tracking the project and completion photos live via their Job Number.

2.4 Operating Environment

The system runs in modern web browsers on desktop and mobile-responsive views; field execution is performed through the responsive web portal. It uses a backend server, relational database, object storage, and Redis cache; all communication via HTTPS/TLS 1.3.

2.5 Design and Implementation Constraints

- Multi-tenant data isolation enforced at the query level.
- ZATCA-compliant 15% VAT partitioning for taxable items.
- Architecture must be integration-ready for future Civil Defense / government portals (RESTful/GraphQL + OpenAPI).
- Field workflows delivered via responsive web (no native mobile app in the current MVP).

3. External Interface Requirements

3.1 User Interfaces

Responsive web UI for desktop and mobile-responsive browser views; tenant-branded portals with dynamic colors and logos injected on authentication. Field forms use HTML5 browser capabilities (camera capture, canvas signature pad).

3.2 Software Interfaces

Integration with a transactional Email provider, Google Maps for facility geolocation, object storage (AWS S3 / GCP Storage), and Redis Pub/Sub for real-time events.

3.3 Communications Interfaces

All traffic over HTTPS/TLS 1.3; WebSocket channels for low-latency chat; real-time webhooks for obstacle/status broadcasting.

4. System Features

4.1 Tenant & SaaS Management

ID	Name	Description	Actor	Priority
FR-T EN-O 1	Multi-tenant Database	Set up the shared corporate database, add a tenant_id foreign key to all dynamic tables, and create migrations that auto-generate	System	High

		tables when a new company registers.		
FR-T EN-O 2	Request Isolation Middleware	Capture tenant_id automatically from requests, inject a global query scope (WHERE tenant_id = current_tenant), and unit-test that cross-tenant access is blocked.	System	High
FR-T EN-O 3	Custom Subdomain Router	Parse the URL to identify the corporate subdomain, map it to the tenant record, and block special characters or reserved keywords (admin, api, main).	System	High
FR-T EN-O 4	Wildcard Domain Config	Configure the web server for wildcard domains (*.yourdomain.com) and expose a real-time subdomain-availability endpoint during sign-up.	DevOps	High
FR-T EN-O 5	Super Admin Matrix Dashboard	Design and build a high-density monitoring dashboard mapping tenants to resource tiers, showing	Super Admin	High

		subscription status, user counts, and storage usage.		
FR-T EN-O 6	Super Admin Control Endpoints	Provide a one-click Suspend/Activate toggle for non-payment, manage tiers (Trial, Basic, Professional), and enforce per-tier limits on concurrent projects and personnel.	Super Admin	High
FR-T EN-O 7	Tenant Branding	Upload logos/favicons with extension (.png, .svg) and size validation, pick Primary/Secondary colors with live preview, and persist hex codes to the tenant config.	Company Admin	High
FR-T EN-O 8	Dynamic Branding Pipeline	Inject tenant colors into CSS variables / Tailwind on authentication so branding renders across buttons, sidebars, and portals.	System	Medium
FR-T EN-O 9	Watermarked PDF Reports	Embed tenant logo and brand colors into exported PDF headers/footers	System	Medium

		and run end-to-end rendering tests for unified presentation.		
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4.2 Client Portal & Intake Lifecycles

ID	Name	Description	Actor	Priority
FR-IN T-01	Intake Form Wizard	Design and build a wizard capturing CR/national-ID 700 numbers, integer built-up area (rejecting non-numeric), and an interactive Google Maps pin extracting latitude/longitude.	Client	High
FR-IN T-02	Facility Ingestion Endpoints	Validate exact 10-digit CR/700 numbers, reject negative or textual area values, and store extracted coordinates in the facility row.	System	High
FR-IN T-03	Legacy Contract Dates	Capture legacy contract start/end dates and mandatory site-coordinator name and direct cellular phone number.	Client	High

FR-IN T-04	Legacy Contract Email Alerts	Store contract timelines with cron alerts 30 days before expiry and email the site coordinator an activation link securing approval for field access.	System	High
FR-IN T-05	Blueprint Dropzone	Build a multi-file drag-and-drop dropzone with progress and instant error states; server-side filter formats (.dwg, .dxf, .pdf, .png), enforce a 50MB cap, and stream uploads to isolated buckets.	Client / System	High
FR-IN T-06	Legal Documentati on Upload	Force upload of Deed of Ownership or Electronic Rental Contracts, classify structure lifecycle (Active, Under Construction, Renovation), and capture Municipality Building Permits.	Client	High

FR-IN T-07	Safety Evaluation Metadata	Process ownership/rental metadata and bind classification state to append descriptive metadata to the consulting engineer's review layout.	System	Medium
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4.3 Rule & Routing Engine

ID	Name	Description	Actor	Priority
FR-RU L-01	Small-Scale Routing	If area < 150 m ² , tag as Fast-Track / Basic Safety Hardware, route to the customer-service queue for a quote, and permit instant report generation upon payment.	System	High
FR-RU L-02	Medium-Scale Routing	If area is ≥ 150 and ≤ 1000 m ² , categorize under Maintenance Strategy & Mandatory On-Site Evaluation and	System	High

		push to the sales CRM pipeline for a site audit.		
FR-RU L-03	Large-Scale Routing	If area > 1000 m ² , categorize under Complex Blueprint Integration & Comprehensive Structural Consultation and open an advanced consulting ticket.	System	High
FR-RU L-04	ISIC Activity Matrix	Map ISIC activity codes to safety risk bands; seed Commercial Kitchens, Chemical Warehousing, Compressed Gas Outlets, and Heavy Workshops as High-Hazard by default.	System	High
FR-RU L-05	High-Hazard Enforcer	If activity maps to High-Hazard, sever fast-track paths regardless of area and force-route to Comprehensive Engineering Review with mandatory legal alerts.	System	High
FR-RU L-06	Low-Risk Bypass	Capture simple safe structures (retail boutiques, offices) not in the hazard matrix and unlock automated technical safety certificate workflows.	System	Medium

4.4 Consultant Portal & Reports

ID	Name	Description	Actor	Priority
FR-CON-01	Engineering Workspace	Design and build a high-density dashboard combining blueprint canvas, geospatial position, and intake data on one screen, with an in-browser viewer and Approve/Return actions.	Consulting Engineer	High
FR-CON-02	Disposition Endpoints	Handle layout disposition updates (Approved / Modification Required) and block a modification return unless structural correction reasons are filled.	System	High
FR-CON-03	Rich Text Report Editor	Provide a WYSIWYG editor (Tiptap/CKEditor) with formatting helpers and shortcut buttons for standard engineering terminology, stripping raw HTML.	Consulting Engineer	High

FR-CON-04	Snippet Library	Let senior consultants save reusable text arrays to an enterprise snippet table.	Consulting Engineer	Medium
FR-CON-05	Fast-Track Certificate	Bind facility name, dimensions, and activity into an immutable PDF template and stamp the tenant's verification seal as a watermark, instantly downloadable in the client portal.	System	High
FR-CON-06	Fire Compartment Form	Provide structured fields for safety calculations proving passive fire-containment integrity.	Consulting Engineer	Medium
FR-CON-07	Secure Signature PDF Fusion	Output restricted PDFs that lock text selection/modification and merge digital signatures and corporate stamps tamper-proof into metadata.	System	High

4.5 Operations & Field Execution

ID	Name	Description	Actor	Priority
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FR-OPS-01	Scheduling Calendar	Build a responsive web calendar of site visits, availability, and allocations with drag-and-drop assignment onto field-engineer calendars.	Operations Officer	High
FR-OPS-02	Field Dispatch (Email)	Inject the Google Maps coordinate link into the field engineer's responsive web view and fire in-platform push notifications and transactional Email payloads on ticket assignment.	System	High
FR-OPS-03	Project Provisioning	On Paid/Financially-Confirmed status, auto-create a project for the operations officer split into 3 silos (Alarm, Suppression, Ventilation) and display them.	System	High
FR-OPS-04	Expense & VAT Logic	Capture vendor invoices in a web layout mapped to a subsystem category (rejecting unclassified), split 15% VAT into a ZATCA	Operations Officer / System	High

		refund ledger, with a taxable/exempt toggle.		
FR-OPS-05	Web Camera Capture & EXIF	Use HTML5 wrappers (<code>input capture="environment"</code>) to prioritize live camera capture over gallery picking, compress client-side, and verify EXIF confirms live capture, not stale-library uploads.	Field Technician / System	High
FR-OPS-06	Workforce & Labor Cost	Tag members as W2 Internal or Outsource; zero out internal labor cost, require day-rates for outsource, and capture wages per project.	Operations Officer	High
FR-OPS-07	Irreversible Payout & Audit	Provide a Confirm Final Outsource Labor Settlement action with a strong confirmation modal; log timestamp, IP, and geo-tag to a WORM store.	Operations Officer / System	High

FR-OPS-08	Responsive Obstacle Logging	Let field staff log execution blockers via a web view with mandatory live browser-camera photos; webhooks broadcast updates and structure logs into legal exhibits.	Field Technician / System	High
FR-OPS-09	Recurring Maintenance	Divide annual maintenance contracts into 4 quarters, auto-generate 4 scheduled site visits, and email team leads 7 days prior; inject tickets into the planning calendar.	System	High
FR-OPS-10	Web Signature & Closure	Disable Complete-Ticket until the ink-signed, stamped report is uploaded via a responsive web interface combined with an HTML5 canvas signature pad; bind the snapshot as an unalterable attachment.	Field Technician / System	High

FR-OPS-11	Commissioning Metrics	Force technicians to verify subsystem metrics (alarm decibels, hydraulic pressure, voltage, psi) with proof photos via browser, then route the payload to the consulting engineer.	Field Technician / System	High
ID	Name	Description	Actor	Priority

4.6 Communication & Analytics Engine

ID	Name	Description	Actor	Priority
FR-COM-01	Tracking Code Engine	Generate unique tracking IDs on project creation (e.g., F1-2026-SRV-0432).	System	High
FR-COM-02	Milestone Tracking View	Display a linear milestone component in the client view, bound to real-time backend status logs.	System	High

FR-C OM- 03	WebSocket Chat	Spin up an isolated, archived chat channel per Work Order (client, agent, manager), writing messages to durable legal storage; restrict file transfers to internal chat.	System	High
FR-C OM- 04	Net Profitabilit y Engine	Compute Project Net Margin = Gross Revenue – (Material Expenses + Outsource Labor + Agent Commissions), pulling ledger entries on audit clearance, and display net % with deviations.	System	High
FR-C OM- 05	Executive BI Dashboards	Build interactive charts (Chart.js/Recharts) of month-over-month revenue and expense weights across Alarm/Suppression/Ventilation divisions.	Compan y Admin	High
FR-C OM- 06	Performan ce Filters & SLA	Filter top-performing project types and maintenance windows; compute task durations vs. SLA windows and	System	Mediu m

		maintain a per-technician efficiency scorecard.		
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4.7 Non-Functional & Security Requirements

ID	Name	Description	Actor	Priority
NFR-SEC-01	Data Encryption	Apply server-side AES-256 encryption to blueprints, receipts, and wage documents before object-store writes.	System	High
NFR-SEC-02	Presigned Links	Disable public bucket reads and serve expiring presigned URLs that lapse 120 seconds after creation.	System	High
NFR-SEC-03	Transport Security	Enforce HTTPS/TLS 1.3 across all application networks.	DevOps	High
NFR-SEC-04	RBAC	Enforce fixed roles (Super Admin, Company Admin, Engineer, Field Technician, Accountant, Client), verify tokens on all routes, and drop profitability/billing payloads from	System	High

		field-worker responses; hide management paths on the frontend.		
NFR-SEC-05	Multi-Factor Auth (TOTP)	Provide QR enrollment with backup seeds and a 6-digit challenge; integrate TOTP (Google/Microsoft Authenticator) and mandate it for admin, operations, and cash accounts.	System	High
NFR-PER-01	Storage Indexing	Index high-utilization columns (tenant_id, job_number, status, created_at) and eliminate N+1 query anomalies with eager loading.	Backend	High
NFR-PER-02	Redis Cache	Cache user state and short-lived OTP tokens; use Redis Pub/Sub for under-2-second event updates.	DevOps	High
NFR-PER-03	Load Testing	Run headless load tests (k6/JMeter/Locust) simulating 3,000 concurrent connections, link to APM, and resolve bottlenecks to preserve stability.	DevOps	High

NFR-INT-01	API Integratio n	Refactor endpoints to strict REST/GraphQL, standardize property naming for government entities, and build a translation adapter for government-portal compatibility.	Back End	Mediu m
NFR-INT-02	OpenAPI Docs	Auto-compile interactive OpenAPI/Swagger documentation for municipal technical integrations.	DevOp s	Mediu m

5. Use Cases

UC-01 — New Licensing Request (Automated Routing)

Actors: Primary — Client; Secondary — Rule Engine, Consulting Engineer **Pre-conditions:**

Client has accessed the tenant portal via the custom subdomain; the system is online.

Main Flow:

1. Client opens the company portal via the custom subdomain.
2. Client fills the intake wizard: activity name and type, ISIC code, CR/700 number (10 digits), actual built-up area (m²), land plot number and location description, GPS coordinates, current safety tools, and uploads (building permit or deed).
3. Client clicks "Submit Request."
4. System reads the Area field and Activity field and routes automatically:
 - Area < 150 m² → Fast-Track / Basic Safety Hardware → customer-service queue for an instant quote and immediate technical report on payment.

- 150–1000 m² → Mandatory periodic maintenance contract → schedule a field inspection and push to the sales pipeline.
 - 1000 m² → Full engineering blueprint execution → open a consulting ticket.
5. If the activity is High-Hazard (buffet/kitchen, gas extensions, flammable-materials workshop), the system severs fast-track regardless of area and forces a non-instant report under Comprehensive Engineering Review.
 6. System creates the ticket and generates a Job Number with a live tracking timeline.

Post-Conditions: Request is classified, routed to the correct track, and visible to the client via Job Number.

Alternate Flows:

- **AF-1 Invalid CR/700:** Number isn't exactly 10 digits → show validation error → client corrects and resubmits.
- **AF-2 Invalid Area:** Negative or textual value → reject with field error.

Exceptions:

- **EX-1 Blueprint Upload Rejected:** Wrong format or > 50MB → reject server-side with descriptive error; no partial storage.

UC-02 — Login & Multi-Factor Authentication

Actors: Super Admin, Company Admin, Consulting Engineer, Operations Officer, Sales Agent, Client
Pre-conditions: User account exists; user is on the login page.

Main Flow:

1. User enters email/phone and password.
2. System verifies the password via secure hash comparison.
3. System checks account status (active, not suspended).
4. If the account is flagged for mandatory MFA (admin, operations, cash), system presents the 6-digit TOTP challenge.
5. System validates the TOTP and creates a session/JWT.
6. System applies RBAC and redirects to the appropriate workspace.

Post-Conditions: User is authenticated within their role scope; field-worker responses exclude profitability/billing payloads.

Alternate Flows:

- **AF-1 Invalid Credentials:** Show "Invalid email/phone or password"; allow retry or password reset.
- **AF-2 Suspended Account:** Deny access with a support message.
- **AF-3 MFA Failure:** Reject the token; allow re-entry or backup-seed recovery.

UC-03 — Field Execution & Project Closure

Actors: Primary — Operations & Projects Officer, Field Technician; Secondary — Consulting Engineer
Pre-conditions: Quote is approved and first payment confirmed; a project was auto-provisioned into 3 silos.

Main Flow:

1. The project appears in the operations officer's web workspace, split into Alarm, Suppression, and Ventilation silos.

2. Officer records procurement: enters supplier name, purchase item (Alarm/Suppression/Ventilation), and invoice value.
3. Officer sets invoice type:
 - Taxable → system forces entry of pre-tax value and the 15% tax amount in an isolated ZATCA-refund field.
 - Non-taxable → tax field is zeroed and disabled.
4. System blocks saving any field invoice without a live browser-camera photo of the original paper invoice (EXIF-verified).
5. Officer enters field labor: name, field role, and legal category — Internal (cost zeroed) or Outsource (mandatory agreed wage entry).
6. On cash payment to outsource labor, officer clicks "Confirm Final Outsource Labor Settlement"; system logs timestamp, IP, and geo-tag to a WORM store.
7. Officer uploads live installation photos that appear instantly in the client portal and admin panel.
8. On obstacles, officer logs a description with live photos via the web view and clicks "Broadcast"; webhooks alert management and update the client tracker and chat room.
9. After installation, officer runs a full operational test, documents commissioning metrics with proof photos, and sets status to "Ready for Final Inspection."
10. To close: the client signs and stamps the inspection report by hand; the officer uploads it via the responsive web interface (with an HTML5 canvas signature pad option) as the mandatory closure lock; the snapshot binds as an unalterable attachment.
11. System notifies the consulting engineer to approve the official technical reports toward certificate issuance.

Post-Conditions: Project is closed with legal proof of execution; net margin is recomputed from validated expenses.

Alternate Flows:

- **AF-1 Missing Subsystem Classification:** Expense submission rejected to protect gross-margin calculations.
- **AF-2 Stale Receipt Photo:** EXIF check fails (library upload, not live) → reject.
- **AF-3 Closure Without Signed Upload:** Complete-Ticket stays disabled until the signed/stamped document is uploaded.

Exceptions:

- **EX-1 Upload Service Error:** Network/storage failure during photo upload → show retry message, preserve form data, do not mark step complete.